1.2.4 Profiting from low costs for batch processing infrastructure

Nick is a data scientist who needs to process massive amounts of measurement data collected from gas turbines. He needs to generate a daily report containing the maintenance condition of hundreds of turbines. Therefore, his team needs a computing infrastructure to analyze the newly arrived data once a day. Batch jobs are run on a schedule and store aggregated results in a database. A business intelligence (BI) tool is used to generate reports based on the data stored in the database.

Because the budget for computing infrastructure is very small, Nick and his team have been looking for a cost effective solution to analyze their data. He finds the following ways to make clever use of AWS's price model:

- *AWS bills virtual machines per second with a minimum of 60 seconds.* So Nick launches a virtual machine when starting a batch job and terminates it immediately after the job finishes. Doing so allows him to pay for computing infrastructure only when actually using it. This is a big game changer compared to the traditional data center where Nick had to pay a monthly fee for each machine, no matter how much it was used.
- *AWS offers spare capacity in their data centers at a substantial discount.* It is not important for Nick to run a batch job at a specific time. He can wait to execute a batch job until there is enough spare capacity available, so AWS offers him a virtual machine with a discount of 75%.